

Physics-Based Formula Reference

AI-Enabled Digital Twin for a MALE UAV Aero Piston Engine — SIH26054

Compiled reference for prototype design

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1. Purpose and Scope

This document collects the standard, publicly known engineering and physics equations relevant to building a **digital twin of a MALE (Medium-Altitude Long-Endurance) UAV’s reciprocating (piston) engine and airframe**, for health monitoring, fault prediction, and remaining-useful-life (RUL) estimation — the SIH26054 problem domain.

Every equation here is standard textbook thermodynamics, aerodynamics, control theory, or machine-learning formulation (Heywood’s *Internal Combustion Engine Fundamentals*, Anderson’s *Aircraft Performance and Design*, Raymer’s *Aircraft Design*, standard Kalman-filtering and fatigue-mechanics literature). None of it is proprietary to any airframe, engine programme, or weapon system — it is the general science any team would use to build engine/airframe monitoring software. Where a specific numeric example is useful, generic illustrative values are used rather than any classified or export-controlled figures.

2. Aircraft/UAV Performance — Sizing and Mission Equations

These equations size the airframe and predict range/endurance — the “physics ground truth” your digital twin checks live telemetry against.

2.1 Lift, Drag, and the Drag Polar

$$L = \frac{1}{2}\rho V^2 S C_L \quad D = \frac{1}{2}\rho V^2 S C_D$$

where ρ = air density, V = airspeed, S = wing reference area, C_L = lift coefficient, C_D = drag coefficient.

Parabolic drag polar (standard low-speed approximation):

$$C_D = C_{D_0} + \frac{C_L^2}{\pi e AR}$$

- C_{D_0} = zero-lift (parasite) drag coefficient
- e = Oswald span efficiency factor (typically 0.7–0.9 for UAVs)
- $AR = b^2/S$ = wing aspect ratio (b = wingspan)

2.2 Zero-Lift Drag Build-Up (Component Wetted-Area Method)

$$C_{D_0} = \frac{1}{S} \sum_i C_{f_i} FF_i Q_i S_{wet,i}$$

- C_{f_i} = skin-friction coefficient of component i (function of Reynolds number, laminar/turbulent)
- FF_i = form factor (accounts for pressure drag from shape)
- Q_i = interference factor
- $S_{wet,i}$ = wetted area of component i

Total wetted area is the sum of components:

$$S_{wet,total} = S_{wet,wing} + S_{wet,fus} + S_{wet,VT} + S_{wet,HT} + \dots$$

Reynolds number (for skin-friction and boundary-layer transition):

$$Re = \frac{\rho V l}{\mu} = \frac{V l}{\nu}$$

l = characteristic length, μ = dynamic viscosity, ν = kinematic viscosity.

2.3 Thrust/Power Required and Available

$$P_{req} = D \cdot V = \frac{1}{2}\rho V^3 S C_D$$

$$T_{req} = D$$

For a propeller-driven MALE UAV, shaft power from the engine converts to propulsive (thrust) power through propeller efficiency η_p :

$$P_{avail} = \eta_p P_{shaft}$$

2.4 Endurance and Range — Breguet Equations

For a **propeller-driven** aircraft (piston engine + propeller — the relevant case here), endurance E and range R in terms of brake-specific fuel consumption c_p (fuel mass per unit power per unit time), propeller efficiency η_p , and the lift-to-drag ratio L/D :

$$E = \frac{\eta_p}{c_p} \cdot \frac{C_L}{C_D} \cdot \ln\left(\frac{W_0}{W_1}\right)$$

$$R = \frac{\eta_p}{c_p} \cdot \frac{C_L^{1/2}}{C_D} \cdot \frac{2}{\sqrt{2\rho S}} (\sqrt{W_0} - \sqrt{W_1}) \quad (\text{constant-}C_L \text{ climb form})$$

Simplified constant-speed/altitude range form:

$$R = \frac{\eta_p}{c_p} \cdot \frac{L}{D} \cdot \ln\left(\frac{W_0}{W_1}\right)$$

- W_0 = initial (takeoff) weight, W_1 = final weight (after fuel burn)
- $(L/D)_{max}$ occurs at the speed where $C_{D_0} = C_L^2/(\pi e AR)$, i.e. parasite drag equals induced drag

Maximum L/D :

$$\left(\frac{L}{D}\right)_{max} = \frac{1}{2} \sqrt{\frac{\pi e AR}{C_{D_0}}}$$

2.5 Stall Speed and Flight Envelope

$$V_{stall} = \sqrt{\frac{2W}{\rho S C_{L,max}}}$$

2.6 Rate of Climb and Service Ceiling

$$RC = \frac{P_{avail} - P_{req}}{W}$$

Service ceiling is conventionally the altitude where RC falls to a small threshold (e.g. 0.5 m/s or 100 ft/min), found by evaluating $RC(h)$ as air density $\rho(h)$ falls with altitude (see §2.7).

2.7 Standard Atmosphere (density vs. altitude)

For the troposphere (ISA model, up to 11 km):

$$T(h) = T_0 - Lh, \quad p(h) = p_0 \left(\frac{T(h)}{T_0}\right)^{\frac{g}{L R_{air}}}, \quad \rho(h) = \frac{p(h)}{R_{air} T(h)}$$

- $T_0 = 288.15$ K, $p_0 = 101325$ Pa, $L = 0.0065$ K/m, $g = 9.80665$ m/s², $R_{air} = 287.05$ J/(kg · K)

This is the model your digital twin uses to convert altitude telemetry into the local air density that feeds every drag, power, and engine thermodynamic equation below — it's the backbone of “individualized, live-condition performance-envelope” logic.

3. Reciprocating (Piston) Engine Thermodynamics

This is the core physics for the **Mean Value Engine Model (MVEM)** that a digital twin runs in real time against telemetry.

3.1 Otto-Cycle Ideal Efficiency (spark-ignition baseline)

$$\eta_{th} = 1 - \frac{1}{r^{\gamma-1}}$$

r = compression ratio, $\gamma = c_p/c_v$ (ratio of specific heats, 1.3–1.35 for combustion gas).

3.2 Diesel-Cycle Ideal Efficiency (compression-ignition, relevant to

turbodiesel MALE engines)

$$\eta_{th} = 1 - \frac{1}{r^{\gamma-1}} \cdot \frac{r_c^\gamma - 1}{\gamma(r_c - 1)}$$

r_c = cutoff ratio (V_3/V_2 , volume at end of combustion / volume at start of combustion).

3.3 Indicated and Brake Work / Mean Effective Pressure

$$IMEP = \frac{W_i}{V_d}, \quad BMEP = \frac{W_b}{V_d} = \frac{P_b \cdot n_R}{V_d N}$$

- V_d = displaced volume per cycle
- W_i, W_b = indicated / brake work per cycle
- P_b = brake power, N = engine speed (rev/s), n_R = revolutions per power stroke (2 for 4-stroke)

Brake power:

$$P_b = \frac{2\pi N T_{torque}}{n_R} \quad (4\text{-stroke: } P_b = \tau_{torque} \omega \text{ directly for a given crank speed})$$

More directly: $P_b = \tau \cdot \omega$, where τ = crankshaft torque and ω = angular speed (rad/s).

3.4 Mechanical, Volumetric, and Fuel-Conversion Efficiency

$$\eta_m = \frac{P_b}{P_i} = \frac{BMEP}{IMEP}$$

$$\eta_v = \frac{\dot{m}_{a,actual}}{\rho_{a,inlet} V_d N / n_R}$$

$$\eta_f = \frac{P_b}{\dot{m}_f Q_{HV}}$$

- $\dot{m}_a$ = air mass flow, $\dot{m}_f$ = fuel mass flow, Q_{HV} = fuel lower heating value.

3.5 Brake Specific Fuel Consumption (BSFC)

$$BSFC = \frac{\dot{m}_f}{P_b}$$

This directly feeds the Breguet range/endurance equations in §2.4 and is one of the most important live-tracked parameters for a health-monitoring digital twin (rising BSFC at constant power = developing inefficiency).

3.6 Air-Fuel Ratio and Equivalence Ratio

$$AFR = \frac{\dot{m}_a}{\dot{m}_f}, \quad \phi = \frac{(AFR)_{stoich}}{AFR}$$

$\phi > 1$ = rich, $\phi < 1$ = lean. Deviation trend is a standard misfire / injector-fault indicator.

3.7 Woschni Correlation — In-Cylinder Convective Heat Transfer

The dominant semi-empirical model used inside MVEM/thermodynamic digital twins to compute cylinder wall heat loss:

$$h = 3.26 D^{-0.2} p^{0.8} T^{-0.55} w^{0.8}$$

- h = heat transfer coefficient (W/m²K), D = cylinder bore (m), p = cylinder pressure (kPa), T = cylinder gas temperature (K)
- w = characteristic gas velocity, modeled as:

$$w = C_1 \bar{S}_p + C_2 \frac{V_d T_r}{p_r V_r} (p - p_{mot})$$

where $\bar{S}_p$ = mean piston speed, T_r, p_r, V_r = reference (e.g. IVC) conditions, p_{mot} = motored (no-combustion) pressure at the same crank angle, C_1, C_2 = empirical constants (combustion vs. gas-exchange phase).

3.8 In-Cylinder Pressure / Crank-Angle Energy Balance (first-law, MVEM core)

$$\frac{dU}{d\theta} = \frac{dQ_{comb}}{d\theta} - \frac{dQ_{ht}}{d\theta} - p \frac{dV}{d\theta} + \sum_j \dot{m}_j h_j$$

This crank-angle-resolved first-law energy balance — combustion heat release minus wall heat transfer minus piston work, plus flow enthalpy through valves — is the differential equation a real-time MVEM integrates; Woschni (§3.7) supplies the $dQ_{ht}/d\theta$ term and a Wiebe function (below) supplies $dQ_{comb}/d\theta$.

3.9 Wiebe Function — Heat Release Rate Shape

$$x_b(\theta) = 1 - \exp \left[-a \left(\frac{\theta - \theta_0}{\Delta\theta} \right)^{m+1} \right]$$

x_b = mass fraction burned, θ_0 = start of combustion, $\Delta\theta$ = combustion duration, a, m = shape parameters (typically $a \approx 5$, $m \approx 2-3$).

3.10 Turbocharger Boost (relevant — Archer-NG-class engines are turbocharged)

Compressor pressure ratio and corrected mass flow (standard turbomachinery similarity form used in engine-model compressor maps):

$$\pi_c = \frac{p_{02}}{p_{01}}, \quad \dot{m}_{corr} = \frac{\dot{m} \sqrt{T_{01}/T_{ref}}}{p_{01}/p_{ref}}$$

Isentropic compressor work / temperature rise:

$$T_{02,s} - T_{01} = T_{01} \left(\pi_c^{\frac{\gamma-1}{\gamma}} - 1 \right), \quad \Delta T_{actual} = \frac{T_{02,s} - T_{01}}{\eta_c}$$

4. Sensor Fusion and State Estimation

Standard estimators used to fuse noisy multi-sensor telemetry (RPM, CHT, EGT, pressure, vibration, IMU, GPS) into a coherent internal state — the “UKF/ESKF” layer referenced across MALE UAV digital-twin literature.

4.1 Discrete-Time State-Space Model

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}, \quad z_k = h(x_k) + v_k$$

x_k = state vector, u_k = control input, z_k = measurement, $w_k \sim \mathcal{N}(0, Q)$, $v_k \sim \mathcal{N}(0, R)$ process/measurement noise.

4.2 Linear Kalman Filter (baseline)

Predict:

$$\hat{x}_k^- = A\hat{x}_{k-1} + Bu_{k-1}, \quad P_k^- = AP_{k-1}A^T + Q$$

Update:

$$K_k = P_k^- H^T (HP_k^- H^T + R)^{-1}$$

$$\hat{x}_k = \hat{x}_k^- + K_k (z_k - H\hat{x}_k^-), \quad P_k = (I - K_k H)P_k^-$$

4.3 Extended Kalman Filter (EKF) — for nonlinear engine dynamics

Same structure as §4.2 but A, H are replaced by Jacobians of the nonlinear f, h evaluated at the current state estimate:

$$A_k = \left. \frac{\partial f}{\partial x} \right|_{\hat{x}_{k-1}^-}, \quad H_k = \left. \frac{\partial h}{\partial x} \right|_{\hat{x}_k^-}$$

4.4 Unscented Kalman Filter (UKF) — preferred for strongly nonlinear

thermodynamic states

Sigma points (n = state dimension, λ = scaling parameter):

$$\mathcal{X}_0 = \hat{x}, \quad \mathcal{X}_i = \hat{x} + \left(\sqrt{(n+\lambda)P} \right)_i, \quad \mathcal{X}_{i+n} = \hat{x} - \left(\sqrt{(n+\lambda)P} \right)_i$$

Weights for mean/covariance recombination:

$$W_0^{(m)} = \frac{\lambda}{n+\lambda}, \quad W_0^{(c)} = \frac{\lambda}{n+\lambda} + (1 - \alpha^2 + \beta), \quad W_i^{(m)} = W_i^{(c)} = \frac{1}{2(n+\lambda)}$$

Each sigma point is propagated through the true nonlinear $f(\cdot)$, then recombined — this avoids linearization error and is standard for UKF-based engine/airframe digital twins.

4.5 Error-State Kalman Filter (ESKF) — common for INS/GPS fusion

Nominal state propagates through full nonlinear kinematics; a small *error state* δx (attitude error, bias errors) is estimated linearly:

$$\delta x_k^- = F_{k-1} \delta x_{k-1}, \quad \delta x_k = \delta x_k^- + K_k (z_k - H \delta x_k^-)$$

then the error state is injected back into the nominal state and reset to zero — standard for fusing IMU + GPS on a UAV.

4.6 GPS/Motion-Model Cross-Check (anti-spoofing logic)

A simple, physically-grounded consistency check: compare GPS-reported velocity/position against the digital twin's own independently predicted kinematic state, and flag divergence beyond sensor-noise bounds:

$$\|z_{GPS} - H\hat{x}_k^-\| > \kappa \sqrt{\text{diag}(S_k)} \Rightarrow \text{flag}$$

where $S_k = HP_k^-H^T + R$ is the innovation covariance and κ is a chosen confidence multiplier (e.g. 3 for $\sim 99.7\%$ under Gaussian assumptions).

5. Fault Prediction, RUL, and Physics-Informed ML

5.1 Paris' Law — Fatigue Crack Growth (feeds acoustic-emission / crankshaft-crack prediction)

$$\frac{da}{dN} = C (\Delta K)^m$$

- a = crack length, N = load cycles, ΔK = stress-intensity factor range, C, m = material constants (empirical, from fatigue testing).

Stress intensity factor range for a simple through-thickness crack:

$$\Delta K = Y \Delta \sigma \sqrt{\pi a}$$

Y = geometry correction factor, $\Delta \sigma$ = applied stress range.

Integrating Paris' Law from initial crack size a_0 to critical size a_c gives cycles-to-failure — the physical basis for RUL estimation from a detected crack-initiation signal.

5.2 Remaining Useful Life (RUL) — General Formulation

$$RUL_t = t_{EOL} - t$$

where t_{EOL} is the (estimated) time/cycle at which a monitored degradation indicator $D(t)$ crosses a failure threshold D_{th} :

$$RUL_t = \inf\{\tau > 0 : D(t + \tau) \geq D_{th} \mid D(t)\} - t$$

Common parametric degradation model (exponential):

$$D(t) = D_0 + \theta_1 \exp(\theta_2 t) + \epsilon(t)$$

with θ_1, θ_2 estimated online (e.g. via particle filter or Bayesian update) from streaming telemetry — the standard approach behind “RUL estimation” boxes in every digital-twin AI/ML layer.

5.3 Physics-Informed Neural Network (PINN) Loss Function

A PINN embeds governing physics as a soft constraint in the training loss, enabling training with little or no historical failure data:

$$\mathcal{L}_{total} = \mathcal{L}_{data} + \lambda_{phys} \mathcal{L}_{physics} + \lambda_{BC} \mathcal{L}_{BC/IC}$$

$$\mathcal{L}_{data} = \frac{1}{N_d} \sum_{i=1}^{N_d} \|\hat{y}(x_i; \theta) - y_i\|^2$$

$$\mathcal{L}_{physics} = \frac{1}{N_f} \sum_{j=1}^{N_f} \|\mathcal{N}[\hat{y}(x_j; \theta)]\|^2$$

$\mathcal{N}[\cdot]$ is the residual of the governing PDE/ODE evaluated using automatic differentiation of the network output — for example the crank-angle energy balance (§3.8) or a simplified Navier–Stokes / heat-equation residual for thermal-fault prediction, or Paris' Law (§5.1) for fatigue-crack PINNs. $\lambda_{phys}, \lambda_{BC}$ are weighting hyperparameters.

5.4 Bayesian / Variational PINN — Uncertainty Quantification

Instead of point-estimate weights θ , a distribution $q_\phi(\theta)$ is learned by minimizing the evidence lower bound (ELBO):

$$\mathcal{ELBO}(\phi) = \mathbb{E}_{q_\phi(\theta)} [\log p(\mathcal{D} | \theta)] - D_{KL}(q_\phi(\theta) \| p(\theta))$$

Predictive uncertainty is obtained by Monte Carlo sampling $\theta^{(s)} \sim q_\phi(\theta)$ and propagating each sample through the network to get an output distribution — this is the standard mechanism behind “B-PINN with variational-inference uncertainty quantification.”

5.5 Motor/Alternator Current Signature Analysis (MCSA) — for the

electrical-signature misfire-detection channel

Fault sidebands appear in the current spectrum at:

$$f_{sb} = f_e (1 \pm 2ks/p)$$

for induction-type analysis (f_e = electrical supply frequency, s = slip, p = pole-pair number, k = harmonic index); for an alternator driven by a reciprocating engine, the more directly relevant relation is that torque ripple at the engine’s firing frequency

$$f_{fire} = \frac{N \text{ (cylinders firing per rev)}}{n_R} \cdot \frac{RPM}{60}$$

modulates the alternator’s output ripple, so a Fourier-domain peak at f_{fire} (or its harmonics) with growing amplitude is the misfire indicator, extracted via FFT of the alternator output current/voltage:

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$$

5.6 Vibration Spectral Analysis (bulk-vibration baseline, and AE

sensing at higher frequency)

Standard FFT-based order analysis compares vibration energy at multiples of shaft rotational frequency (orders); acoustic-emission (AE) sensing targets ultrasonic stress-wave events (20 kHz – 1 MHz), captured via the same FFT/PSD (power spectral density) approach applied to a much higher-bandwidth signal:

$$PSD(f) = \lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} [|X_T(f)|^2]$$

5.7 Common-Cause / Redundancy-Integrity Monitoring

A simple runtime statistical test for hidden shared-failure coupling between two nominally independent channels A, B : monitor their cross-correlation coefficient over a sliding window and flag when it exceeds an expected-independence bound:

$$\rho_{AB} = \frac{\text{Cov}(A, B)}{\sigma_A \sigma_B}, \quad |\rho_{AB}| > \rho_{th} \Rightarrow \text{flag "redundancy integrity degraded"}$$

6. Control / Decision Layer

6.1 Model Predictive Control (MPC) — Edge-Native Fault Mitigation

At each step, solve a finite-horizon optimal control problem over the digital twin’s predictive model:

$$\begin{aligned} \min_{u_0, \dots, u_{H-1}} \quad & \sum_{k=0}^{H-1} \left[\|x_k - x_{ref}\|_Q^2 + \|u_k\|_R^2 \right] + \|x_H - x_{ref}\|_{Q_f}^2 \\ \text{s.t.} \quad & x_{k+1} = f(x_k, u_k), \quad x_k \in \mathcal{X}, \quad u_k \in \mathcal{U} \end{aligned}$$

Only the first control action u_0^* is applied, then the horizon is re-solved at the next timestep (receding-horizon) — this is the structure behind “edge-native embedded MPC” for millisecond throttle/mixture correction.

6.2 Glide-Footprint / Best-Reachable-Site Calculation

For an engine-out or power-reduced condition, reachable ground range from current altitude h and best glide $(L/D)_{max}$ (still air):

$$R_{glide} \approx h \cdot \left(\frac{L}{D} \right)_{max}$$

Corrected for wind component V_w along a given bearing and glide airspeed V_g :

$$R_{glide,wind} \approx h \cdot \left(\frac{L}{D} \right)_{max} \cdot \left(1 + \frac{V_w}{V_g} \right)$$

This generates the reachable-landing-site circle/ellipse used for autonomous divert planning and emergency recovery.

7. Suggested Symbol Table

Symbol	Meaning	Typical Units
ρ	Air density	kg/m ³
V	Airspeed	m/s
S	Wing reference area	m ²
C_L, C_D	Lift / drag coefficient	—
AR	Aspect ratio	—
e	Oswald efficiency factor	—
r	Compression ratio	—
γ	Ratio of specific heats	—
$BMEP, IMEP$	Brake / indicated mean effective pressure	Pa
$BSFC$	Brake specific fuel consumption	kg/(kW · h)
η_v, η_m, η_f	Volumetric / mechanical / fuel-conversion efficiency	—
h	Convective heat transfer coefficient (Woschni)	W/(m ² K)
a	Crack length (Paris' Law)	m
ΔK	Stress intensity factor range	MPa · $\sqrt{m}$
$D(t)$	Degradation indicator	(indicator-specific)
x_k, P_k	Kalman filter state / covariance	(state-specific)
θ	Neural-network weights	—

8. Notes on Use

- These are the **standard, open-literature equations** used to build a physics-informed digital twin (MVEM thermodynamics + Woschni heat transfer + Wiebe combustion model is the textbook combination for real-time engine simulation; UKF/ESKF is the textbook combination for UAV state estimation; Paris' Law + PINN is the textbook combination for physics-informed fatigue/RUL prediction).
- Constants such as C, m (Paris' Law), a, m (Wiebe), C_1, C_2 (Woschni), and Oswald efficiency e are **empirical and material/engine-specific** — they must be calibrated against real test data or literature values for your specific engine and airframe, not assumed.
- None of these equations are specific to weapon systems, targeting, or munitions — they describe engine thermodynamics, aerodynamics, sensor fusion, and fault prognosis, the same physics used in any predictive maintenance system (automotive, marine, industrial turbine, or civil aircraft).